Fiscal Year:	FY 2012	**Task Last Updated:**	FY 08/22/2012
PI Name:	Mednieks, Maija Ph.D.		
Project Title:	Protein Expression in Salivary Glands: Effects of Extended Space Flight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
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City:	Farmington	**State:**	CT
Zip Code:	06030-0001	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	07/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
Contact Email:	Sidney.C.Sun@nasa.gov		
Flight Program:	Bion/Shuttle		
Flight Assignment:	STS-131 ; STS-133 ; Bion M1 NOTE: Extended to 7/31/14 per A. Chu/ARC and NSSC info (Ed., 4/30/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hand, Arthur D.D.S. (University of Connecticut Health Center)		
Grant/Contract No.:	NNX09AP13G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The proposed study is to determine effects of extended space travel on oral tissues using the salivary glands of mice. Proteins secreted from the salivary glands play an important role in maintaining the integrity of the mouth and teeth. Rodent salivary glands are structurally and functionally similar to those of humans therefore useful for research on changes associated with travel in space. Namely, studying the effects of zero gravity (ØG) on tissue, cellular and molecular levels. Our previous studies on short duration (~ 1 week) space flight missions showed that specific secretory proteins were modified by exposure to both zero and hyper - G conditions. Objective: to test the hypothesis that a long duration (~ 1 month) space flight will show effects of extended ØG on protein expression in salivary glands of the experimental animals. Methods: salivary glands from flight animals and ground-based controls will be prepared for biochemical (electrophoresis and Western and Northern blotting) and morphological (electron microscopy and immunocytochemistry) experiments to determine changes in ultrastructure and in expression and intracellular localization of proteins that are markers for different cell types in the salivary glands or are indicators of specific cell functions. Several of the proteins have previously been shown to respond to a variety of environmental stimuli, including space flight. Significance: proteins present in secretion granules of salivary glands are released into saliva and can be used to monitor effects of environmental stress. Employing saliva as a diagnostic test medium could have clinical applications and be useful in measuring human responses to space flight. Conclusions: measuring and quantification of proteins in salivary glands will show if space travel affects the physiology of the oral cavity as well as the regulation of cellular signaling and molecular signal transduction of secretory cells.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Background and Objectives: It is important to study the effects of travel in space and to measure reactions to the stress of weightlessness in zero gravity (ØG) conditions. Stress changes hormone action that is reflected in different body parts/organs as well as in glands that produce saliva. The goal is to devise a clinical test for stress disorders in space and on Earth. Saliva contains substances (e.g. proteins) that change during travel in space. We can determine which salivary proteins have been changed by using biochemical tests and histologic analyses to study tissues and cells (salivary glands) that produce saliva in mice flown on the space shuttle Discovery STS-131 and 133 missions. This information we will be used to design a test kit to measure responses in astronauts so no blood will have to be drawn. Collecting saliva instead of blood and urine, will be easier during space flight and will make it possible to study hormone changes experienced by Astronauts and Cosmonauts at ØG. Travel in space is an ideal model system for studying a specific environmental impact on physiologic function that is due to gravitational change and cannot be duplicated on earth. Understanding the effect of ØG will define changes due to the environment as the stressor. This information can then be adapted to devising specific tests of different conditions using saliva for general clinical applications in medicine in space or on Earth. Research Description: The Biospecimen Sharing Program has provided an unparalleled opportunity to study tissue and secretory protein signaling in tissues affected by environmental stimuli whereby cellular and molecular changes associated with travel in space can be identified in experimental animals and can be monitored by measuring the expression of specific proteins. Previous studies on the effects of space flight have shown that regulation by catecholamine hormones is markedly altered during space flight. The expression of specific proteins as markers can potentially be used for measuring responses to environmental stress since changes in the expression of secretory proteins are reflected in saliva.
Task Progress:	The study was designed to determine effects of extended space travel on oral tissues using the salivary glands of mice. Rodent salivary glands are structurally and functionally similar to those of humans therefore useful for research on changes associated with travel in space: namely, the effects of zero gravity (ØG) on tissue, cellular and molecular levels. Proteins secreted from the salivary glands play an important role in maintaining the integrity of the mouth and teeth. Additionally, several of these proteins are markers for physiologic processes such as digestive and immune systems and hormonal regulation. Results of the present study are in agreement with those of studies on the Cosmos missions and between STS-131 and 133. The proteins in saliva are reliable indicators of the environmental effects of travel in space. The physiologic processes for which they are markers are either slowed, accelerated or unaffected. Initial experiments on recovery of salivary protein changes show that by day 5 after landing a number of the protein concentrations have returned to normal. The next experiments will be to use microarray systems to determine which genes are affected. The present results indicate that travel in space alters the expression of specific proteins. Notably, proteins associated with hormonal signaling and gene expression in salivary glands are down-regulated. Other secretory proteins, however, appear increased and still others are unaffected. The observed phenomena are therefore related to function rather than being due to some overall, nonspecific physiologic functional disruption. The responses to ØG by secretory proteins that may be specific for metabolic, signaling pathways or be indicatory of digestive or immune processes can be the basis for designing measuring devices to test for the disruption of these processes in astronauts. Ultimately such tests can be adapted for clinical or field use on Earth.
Bibliography Type:	Description: (Last Updated: 02/04/2021)
Abstracts for Journals and Proceedings	Mednieks M, Tsesis M, Manzi D, Larson N, Hand AR. "Protein Expression in Mouse Salivary Glands: Effects of Spaceflight." 2012 IADR (International Association for Dental Research) Conference, San Diego, CA, March 16-19, 2011. Journal of Dental Research 2011:2372. http://iadr.confex.com/iadr/2011sandiego/webprogramschedule/Paper148357.html , Mar-2011
Abstracts for Journals and Proceedings	Mednieks M, Perrier J, Khatri A, Khan I, Hand AR. "Secretory protein expression under microgravity conditions." 2011 American Society for Cell Biology, Denver, CO, December 7-11, 2011. 2011 American Society for Cell Biology, Denver, CO, December 7-11, 2011. , Dec-2011
Significant Media Coverage	Maher MA. "Article, 'Of Mice and Men,' describing PI's flight experiment." In the Woods Newsletter 2011 May;29(5):1,4. http://www.farmingtonwoods.com/Newsletters.aspx and search within the May 2011 issue., May-2011